AIQG — Patent Candidates: Scout & Autolearning

Status: Working analysis / counsel-ready draft — 2026-06-19. Technical/strategic assessment, **not legal advice**; patentability and FTO calls require a patent attorney. **Inputs:** `aiqg-ui` Experiments (Scout panel), `aiqg-dashboard-be` `/api/v1/metrics/by-vendor` + verdict/significance, `aether-shared/data-models/aiqg/experiment.md`, `CLAIMS_TRACEABILITY.md` (claims C7/C9/C13). **Family companions:** `AIQG-PATENT-ANALYSIS.md`, `AIQG-PROVISIONAL-BRIEF.md`, `AIQG-FTO-CLAIM-READ.md`

Summary

Two adjacent candidates that extend the attribution family into the **experimentation + adaptation** layer:

- **Scout (shipped 2026-06-18)** — a traffic-mining experiment-prioritization engine whose distinguishing feature is a **pre-launch time-to-verdict estimate** computed from per-model cost/latency/quality variance observed in the gateway's own historical traffic.
- **Autolearning (design-only; the open loop in C9/C13)** — closing the feedback loop so that statistically-proven experiment verdicts and human/ implicit feedback **automatically and safely adapt** routing, judge calibration, and guardrails, gated by the inferred identity ladder.

Honest prior-art posture up front: adaptive/bandit model routing and online learning are a **crowded field** (RouteLLM, bandit model selection, A/B auto-promotion). Neither candidate should be filed on the bare idea of "learn to route." The defensible novelty in both is a **specific combination** with AIQG-only primitives — the inferred identity ladder, CLEAR's decomposed scoring, and judge-calibration coefficients — not the generic adaptation loop.

Candidate S1 — Pre-launch time-to-verdict from observed per-model variance (Scout)

Mechanism. Before any experiment is launched, the system: 1. aggregates the gateway's historical traffic per `(vendor, model)` over a window, computing mean and **standard deviation** of cost, latency, and CLEAR quality (`/api/v1/metrics/by-vendor` returns `cost_stddev` / `latency_stddev` per vendor/model); 2. for a proposed incumbent→candidate swap at a default exposure (e.g. 50/50, 5% of traffic), computes the **sample size to statistical significance** from those observed variances — `requests_to_significance ≈ (z / Δmean)² · (σ_a² + σ_b²)` — and converts it to a **time-to-verdict** at the model's observed request rate; 3. **surfaces the estimate before the operator commits** ("≈ N days", or "traffic too thin"), flagging amber when the projected verdict time exceeds a threshold (~90 days).

Why potentially novel. A/B power analysis is textbook; doing it *up front* from **per-model variance harvested from the same live gateway traffic the experiment will run on**, to triage which swaps are even worth testing, is the differentiated step. The system answers "is this experiment feasible, and how long will it cost you?" before a single live request is split — turning power analysis from a post-hoc check into a **prioritization filter**.

Status: shipped (UI `ScoutPanel`, `deriveSuggestions()`; backend by-vendor stddev + significance calc). Per-model variance integration finished via aiqg-dashboard-be#66.

Candidate S2 — Confounding-aware suggestion → auto-drafted controlled experiment (Scout)

Mechanism. Scout mines historical traffic for incumbent→candidate model swaps that achieved **comparable quality at lower cost/latency**, ranks them by ROI, and — crucially — **explicitly models the confounding**: historical numbers are confounded because the two models served *different* requests, so Scout presents each as a *hypothesis only* and **auto-drafts the controlled split experiment** (variants, cohort, guardrails) that would actually prove it, prefilled and one click from launch.

Why potentially novel. The combination of (a) automated traffic-mined candidate discovery, (b) first-class confounding disclosure in the recommendation, and (c) one-click materialization of the suggestion into a properly-powered controlled experiment closes the *observe* → *hypothesize* → *prove* loop inside one tool. Weaker than S1 on its own; strongest claimed as a dependent combination with S1's time-to-verdict gate.

Status: shipped (suggestion engine + Runner draft prefill).

Candidate A1 — Closed-loop verdict-to-routing promotion, gated by the identity ladder

Mechanism (design-only). When the experiment Runner reaches a verdict that a candidate is **statistically non-inferior on quality and a strict win on cost/latency** (existing z-test non-inferiority + objective-win logic), the system **automatically promotes the candidate into routing policy** for the specific cohort that was tested — but only for traffic whose **inferred identity (agent cluster / flow type) and identity_confidence** match the experiment's cohort. The learned route carries provenance (which verdict authorized it) and auto-reverts if live guardrails (error/latency/cost deltas) breach.

Why potentially novel. Auto-promotion of A/B winners exists. The differentiator is **gating the learned routing decision on the zero-cooperation inferred-identity ladder** from the attribution family — the system applies a learned route to *un-instrumented* callers it recognizes by fingerprint cluster, not by a self-asserted tag, with confidence-weighted rollout. This ties A1 directly to provisional Candidates 3–4 and is hard to replicate without that attribution layer.

Status: design-only. Today: verdicts render to a human (C7 shipped); policy is static observe-only (C9); no promotion path exists.

Candidate A2 — Feedback-recalibrated scoring (calibration coefficients as live correction, not display)

Mechanism (design-only). AIQG already ingests explicit/implicit feedback (thumb, accept_reject, rating, task_success, reward, edit_distance) keyed to response events, and already computes **judge-calibration metrics** (agreement, bias, MAE) between the LLM-judge's CLEAR-efficacy score and human outcomes — **but only displays them**. A2 turns those metrics into **live per-(agent cluster, workflow_type) correction coefficients**: the measured judge bias/MAE recalibrates future CLEAR efficacy/assurance scores, and contradicting feedback **suppresses repeat false-positive flags**. The calibration card becomes a control input, not a readout.

Why potentially novel. Closing the judge → calibration → score-correction loop **scoped per inferred agent cluster** (so a judge that's systematically harsh on one agent type is corrected only there) is the differentiated piece. Generic RLHF / judge-calibration is prior art; **per-cluster online correction driven by the inferred-identity scoping** is the AIQG-specific combination.

Status: design-only. Feedback ingest + calibration compute shipped (C13 "open loop"); no consumer applies them.

Candidate A3 — Adaptive guardrails + autonomous next-experiment proposal

Mechanism (design-only). Two pieces: (i) replace today's **hardcoded** auto-stop guardrail thresholds (error delta, latency factor, cost factor) with **thresholds learned per agent cluster** from that cluster's observed variance (the same σ Scout already computes); (ii) feed verdict outcomes + Scout suggestions into a loop that **proposes the next experiment automatically** (autonomous experimentation) within a tenant-set risk budget, never auto-launching without the A1 promotion gate.

Why potentially novel. The combination of variance-derived per-cluster guardrails with Scout's time-to-verdict feasibility filter to drive an auto-proposed experiment queue. Lowest-priority of the three; most exposed to bandit/auto-experimentation prior art. Likely better as a **dependent claim** on A1/S1 than a standalone.

Status: design-only.

What weakens or gates the position

- **Prior-art density (autolearning).** Adaptive model routing, multi-armed bandits over models, and automated A/B promotion are heavily published and patented. A1–A3 must be claimed as **combinations with the inferred-identity ladder + per-cluster CLEAR calibration**, never as generic adaptive routing. A dedicated prior-art / FTO search on *bandit model routing* and *automated experiment promotion* is required before filing A-series.
- **Scout S1 is the stronger, cleaner candidate** — it is shipped (reduction to practice exists), and "up-front time-to-verdict from same-traffic per-model variance" is a narrower, more defensible step than the A-series.
- **FTO not yet read for these.** The existing claim-read (`AIQG-FTO-CLAIM-READ.md`) covered the attribution techniques, not experimentation/adaptation. The four patents there (esp. Dynatrace 10,924,326 on metric attribution) do **not** obviously read on Scout/autolearning, but a fresh search is owed.
- **Autolearning is unbuilt.** A provisional can cover it with a sufficiently enabling description, but value is far higher once A2's feedback→score loop is prototyped against `cmd/demo-traffic` (reduction to practice).

Recommended next steps

1. **Fold Scout S1 into the main provisional** (with Candidates 1–4) — it is shipped, narrow, and shares the gateway/CLEAR substrate. S2 as a dependent.
2. **Hold the A-series (autolearning) for a second provisional** after (a) a bandit/auto-experimentation prior-art search and (b) a prototype of A2's feedback→score-correction loop. Claim them only as combinations with the inferred-identity ladder.
3. **Prototype A2 first** — it is the most defensible (per-cluster judge recalibration), reuses already-shipped feedback + calibration data, and produces the reduction-to-practice artifact that lifts the A-series from idea to filing-ready.

4. **Disclosure hygiene:** keep the autolearning design non-public until the prior-art search settles provisional-vs-defensive-publication for the A-series (same statutory-bar guard as the attribution family).